Assigned: Friday Aug. 28, 2026
Due: Friday Sept. 11, 2026

MATH 6660: Homework 1
Fall 2026

1. Consider the following 2-point BVP:

$$\mathcal{L}(u) := -\big(a(x)\,u'\big)' = f(x) \quad \text{on} \quad 0 < x < 1,$$
$$u(0) = 0, \quad u(1) = 0,$$

where $f \in C^0(0,1)$, $a \in C^1(0,1)$, and $0 < a_1 \le a(x) \le a_2 < \infty$. In class, we showed that for any two continuous functions on $(0,1)$, u and v, that satisfy the boundary conditions, $u(0) = v(0) = u(1) = v(1) = 0$, the operator, $\mathcal{L}$, satisfies Green's formula:

$$\int_0^1 \left[u\mathcal{L}(v) - v\mathcal{L}(u)\right] dx = 0.$$

(a) Consider the associated eigenvalue problem:

$$\mathcal{L}(\phi) = \lambda\,\phi, \tag{1}$$

where λ is an eigenvalue and $\phi(x)$ is the corresponding eigenfunction. Prove that λ must be real. **HINT:** start by showing that if $(\lambda, \phi(x))$ is an eigenvalue/eigenvector pair, then so is $(\overline{\lambda}, \overline{\phi}(x))$, where $\overline{\cdot}$ means complex conjugation.

(b) Prove that two eigenfunctions, ϕ_1 and ϕ_2, corresponding to two different eigenvalues, λ_1 and λ_2, respectively, where $\lambda_1 \neq \lambda_2$, are **orthogonal**:

$$\int_0^1 \phi_1(x)\,\phi_2(x)\,dx = 0.$$

(c) Prove that all the eigenvalues of (1) are strictly positive: $\lambda > 0$. **HINT:** Multiply (1) by ϕ and integrate over $(0,1)$.

(d) Let $a(x) \equiv 1$. Consider the special case where the load function, f, is chosen as the Dirac delta centered at s:

$$\mathcal{L}(G) = -G'' = \delta(x - s), \quad G(0;s) = G(1;s) = 0.$$

Compute by hand the solution to the above problem: $G(x;s)$. This is known as the **Green's function**.

(e) Let $a(x) \equiv 1$. From the previous part, show that $G(x;s) = G(s;x)$. Using an approriate application of Green's formula, show that for a general load function, $f \in C^0(0,1)$, the solution to

$$\mathcal{L}(u) = -u'' = f(x), \quad u(0) = u(1) = 0$$

is given by

$$u(x) = \int_0^1 f(s)\,G(s;x)\,ds.$$

2. Consider the following scalar product and the associated norm:

$$(v,\, w) := \int_0^1 v(x)\,w(x)\,dx \qquad \text{and} \qquad \|v\| := \sqrt{(v,\, v)}.$$

Prove Cauchy's inequality:

$$|\,(v,\,w)\,| \leq \|v\|\|w\|.$$

HINT: Construct the function: $P(t) := \|v - tw\|^2$, and show that P is a quadratic polynomial. Think about what you can say about the roots of P.

3. Consider the following boundary value problem with periodic boundary conditions:

$$\begin{aligned} \textbf{ODE}: &\quad -u'' + q(x)\,u = f(x), \\ \textbf{BC1}: &\quad u(-\pi) = u(\pi), \\ \textbf{BC2}: &\quad u'(-\pi) = u'(\pi), \end{aligned}$$

where $q > 0\ \forall x \in [-\pi, \pi]$.

(a) Recast this equation into a variational problem, stating the trial and test function spaces.

(b) Interpret the variational problem as an energy minimization problem, clearly stating the energy functional. Prove that the variational problem and the energy minimization problems are equivalent.